

# Raj Khatri

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## PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working at the intersection of plant diagnostics and applied GenAI. At Tata Motors, diagnosed a live Body-in-White production defect by separating an upstream supplier-side cause from on-line equipment wear, using regression-based trend analysis on inspection data and confirming it by physical measurement on the shop floor. Built and deployed Source-Governed RAG, a retrieval system in which source approval status overrides semantic similarity — raising retrieval accuracy from 78% to 95% and forcing refusal rather than ungrounded generation when the corpus cannot support an answer. Reads plant process reasoning and writes Python.

## EDUCATION

**B.Tech, Industrial & Production Engineering**  
*College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar*

Uttarakhand, India  
**August 2023 – August 2027**

## TECHNICAL SKILLS

|               |                                                                                                                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages:    | Python, SQL (PostgreSQL / MySQL), JavaScript, Bash                                                                                                                                                                 |
| GenAI & ML:   | RAG pipeline design, agentic tool routing, vector search (ChromaDB, Postgres pgvector), sentence embeddings (MiniLM / ONNX), grounded generation, refusal & decline behaviour, evaluation-set design, Scikit-learn |
| Backend:      | FastAPI, Pydantic data contracts, SQLAlchemy, PostgreSQL, pgvector, star-schema modelling, Flask, Next.js, Supabase, REST API design                                                                               |
| Data & Tools: | Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI, Git/GitHub, Render, Vercel, Neon                                                                                                    |
| Domain:       | Root-cause analysis, upstream/downstream fault isolation, DFMEA, Six Sigma, statistical modeling & hypothesis testing, evaluation-set design                                                                       |

## EXPERIENCE

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| <b>Tata Motors</b><br><i>Industrial Trainee — Data Analytics</i><br><i>Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"</i> <ul style="list-style-type: none"><li>Diagnosed a recurring Body-in-White quality failure as an upstream supplier-side dimensional defect, ruling out on-line equipment wear as the cause.</li><li>Applied linear regression with R<sup>2</sup> confidence scoring across ~300 inspection data points, then validated the result by physical measurement on the shop floor.</li><li>Designed a containment procedure deployable without production stoppage or vendor dependency.</li><li>Built an interactive diagnostic dashboard tracking defect trends across 10 monitored quality metrics for ongoing condition review.</li><li>Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.</li></ul> | Pantnagar, Uttarakhand ( <b>On-site</b> )<br><b>03 July 2026 – 03 August 2026</b> |
| <b>Voldebug Innovations Pvt. Ltd.</b><br><i>Backend Developer (Part-time)</i> <ul style="list-style-type: none"><li>Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts.</li><li>Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer.</li><li>Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.</li></ul>                                                                                                                                                                                                                                                               | Remote<br><b>22 August 2026 – Present</b>                                         |
| <b>Intern</b> <ul style="list-style-type: none"><li>Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system.</li><li>Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>01 July 2026 – 31 July 2026</b>                                                |

## PROJECTS

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| <b>Supplier Quality Risk Agent</b> <i>Independent project · Python, FastAPI, Postgres + pgvector, Pydantic · Render / Vercel / Neon</i> <ul style="list-style-type: none"><li>Built a dual-source AI agent that decides per query whether an answer requires structured supplier data, corrective-action documents, both, or neither — and declines when no source can support one — reaching 88–92% routing accuracy on a 24-question evaluation set.</li><li>Engineered a paired test case in which two suppliers show statistically identical defect-rate increases requiring opposite responses — one a genuine process degradation, one a measurement artefact — making single-source failure demonstrable rather than theoretical.</li><li>Replaced text-to-SQL with a fixed catalogue of eight parameterised queries, trading question coverage for reviewability in a system where a wrong number triggers a supplier corrective action.</li><li>Caught a regression where a prompt improvement made the router fabricate evidence for a non-existent supplier; constrained the prompt with the real entity list, raising decline accuracy to 5/5.</li><li>Committed the routing ground-truth rule and labelled evaluation set to version control before writing the router, so reported accuracy could not be fitted to observed behaviour; reported results as a range across five runs rather than a single best score.</li></ul> |                  |
| <b>Source-Governed RAG</b> <i>Python, FastAPI, ChromaDB, MiniLM/ONNX, PostgreSQL, Next.js · Render / Vercel / Neon</i> <ul style="list-style-type: none"><li>Built an end-to-end retrieval-augmented generation system over regulatory documents, combining vector search, grounded generation, and a governance layer that vetoes untrustworthy sources.</li><li>Improved retrieval accuracy from 78% to 95% on a labelled evaluation set by replacing fixed-size character chunking with document-structure chunking.</li><li>Architected a governance layer where approval status and review dates override similarity ranking, so expired sources cannot answer.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Live demo</b> |